Genotype-Phenotype Mapping in Cystic Fibrosis: A Predictive Model of Disease Severity Based on Allelic Combinations

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***Abstract*—Cystic fibrosis (CF) is a single-gene disorder, yet its severity varies widely, even between patients who share a genotype. Laboratories have now measured the functional impact of many CFTR variants, but how far those measurements, or computational effect scores, actually predict a patient's clinical outcome remains poorly quantified. This work closes that gap with a reproducible pipeline that maps variant-level molecular features onto patient-level phenotypes, using the phenotype summary of Sosnay et al. (159 CFTR variants from roughly 39,696 patients). A multi-output Random Forest, evaluated by leave-one-out cross-validation, predicted mean sweat chloride and pancreatic insufficiency with moderate accuracy ($R^2 = 0.506$ and 0.545), but mean FEV1 was effectively unpredictable from genotype alone ($R^2 \approx -0.04$). We read this as a biological result and not a modelling failure, since lung function is governed largely by modifier genes, infection and environment. Extending the model to diploid genotypes, we show that a naive split of allele pairs inflates sweat-chloride accuracy to $R^2 \approx 0.85$, whereas a leakage-controlled, variant-grouped split gives an honest 0.328, exposing allele memorisation as an artefact. Every finding held across Random Forest, CatBoost and TabPFN, which suggests the results belong to the data and not to one algorithm. The study sets out which CF phenotypes are genotype-predictable, and offers a leakage-aware template for monogenic disease.**

***Index Terms*—Cystic fibrosis, CFTR, genotype-phenotype mapping, Random Forest, multi-output regression, unsupervised clustering, variant effect prediction, data leakage.**

I. Introduction

Cystic fibrosis is one of the most common life-limiting recessive disorders in people of European ancestry, and arises from pathogenic variation in a single gene, the cystic fibrosis transmembrane conductance regulator (CFTR) (Riordan et al., 1989). CFTR sits in the apical membrane of epithelial cells as a cyclic-AMP-regulated anion channel (Gadsby et al., 2006); when it fails, chloride and bicarbonate can no longer cross the epithelium normally, and because this affects many organ systems at once, CF presents as a multi-system disease (Cutting, 2015). More than two thousand CFTR variants have been reported, only a subset clearly disease-causing (CFTR2, 2026). Genotype is nonetheless an imperfect guide to phenotype. Two patients with the same genotype can differ considerably, most visibly in lung function, because modifier genes, environment and chance all play a part (Cutting, 2015). Classifying variants by functional and clinical impact therefore matters directly for diagnosis and for treatment, including who qualifies for the newer variant-specific CFTR modulator therapies (Ramsey et al., 2011).

Two kinds of evidence have recently become abundant. Standardised in-vitro assays now quantify how variants affect chloride conductance, protein processing and RNA splicing (Sosnay et al., 2013), and computational predictors estimate how damaging a substitution is likely to be (Adzhubei et al., 2010; Kumar et al., 2009; Mi et al., 2013). If clinical severity could be read directly from a variant's molecular signature, it would sharpen diagnosis and prognosis and help decide who is likely to benefit from modulator therapy. Yet the field has not established which outcomes are genuinely recoverable from these features, nor how much of the apparent skill of genotype-pair models survives leakage-controlled evaluation (Kaufman et al., 2012). This dissertation addresses that gap directly, and treats the honesty of the evaluation as a first-class concern. The remainder of the dissertation is arranged as follows. Section II states the problem, Section III sets out the contributions, Section IV reviews related work, Section V describes the data and methods, Section VI presents the results, Section VII discusses them, and Section VIII concludes.

II. Problem Statement

Two properties of the data shape the prediction task. First, the clinically useful severity measures in the Sosnay summary are continuous rather than categorical. The table does assign each variant a categorical clinical call (CF-causing, Indeterminate or Not CF-causing), but that call is itself built partly from the same clinical numbers, so it is kept only for colouring plots and for checking the structure the models find, never as something to predict. Regressing the measurements directly avoids squashing a graded biological signal into three boxes. Second, the molecular data describe one allele at a time while a patient carries two. A clinically meaningful model therefore has to fold two allelic descriptions into a single genotype-level prediction. It should not depend on the arbitrary order in which the two alleles happen to be written. It should respect the recessive expectation that residual CFTR function is set by

the milder allele. And its evaluation should never let an allele appear in both training and testing.

III. Set of Contributions

This dissertation makes four contributions. First, a reproducible pipeline that maps variant-level molecular features, meaning the functional assays and in-silico effect scores, onto patient-level continuous phenotypes. Second, a concrete answer to which CF phenotypes are predictable from genotype-level features: sweat chloride and pancreatic insufficiency with moderate accuracy, lung function not at all, a result that survives replacing the Random Forest with gradient-boosted (CatBoost) and transformer-based (TabPFN) models. Third, a worked example of how data leakage arises in genotype-pair modelling and how variant-grouped cross-validation fixes it. Fourth, a clinician-facing artefact: a symmetric table of predicted sweat chloride across pairs of the twenty most frequent CFTR variants, offered as an illustration of the severity gradient and not as a validated predictor.

IV. Related Work

The first attempts to link CFTR genotype to clinical phenotype established broad correlations between variant class and disease severity. These were most reliable for pancreatic status, where variants divided into pancreatic-insufficient and pancreatic-sufficient groups with reasonable consistency (Kerem et al., 1990; Borgo et al., 1990). Sweat chloride, the standard diagnostic biomarker of CFTR channel function (Farrell et al., 2017), behaved similarly and tended to separate severe genotypes from milder ones (Wilschanski et al., 1995). Studies of ΔF508 quantified this in homozygous and compound-heterozygous patients (Johansen et al., 1991), and the same genotype-phenotype logic extends to glucose homeostasis in CF (Preumont et al., 2007) and, beyond CF itself, to CFTR-related disorders such as chronic pancreatitis (Sharer et al., 1998). The dataset used here comes from a multi-centre project that defined the disease liability of CFTR variants by combining clinical, functional and computational evidence (Sosnay et al., 2013). One observation recurs throughout this literature: the correlation is distinctly weaker for lung function than for other outcomes (Cutting, 2015). That unexplained variance has prompted a large body of work on modifier loci outside CFTR itself (Wright et al., 2011; Shanthikumar et al., 2019), and frames the expectation, tested directly here, that lung-function endpoints are less amenable to genotype-based prediction than biomarkers closer to channel function.

Predicting the consequence of a variant by computation is not a new idea. Tools such as PolyPhen-2, SIFT and PANTHER use sequence and structural information to estimate how damaging an amino-acid substitution is likely to be (Adzhubei et al., 2010; Kumar et al., 2009; Mi et al., 2013), but what they return is a generic measure of functional impact, not a clinical outcome that anyone has measured. More recent machine-learning work in CF has taken on pathogenicity classification and prognostic scores whose external validity across registries has itself become a subject of study (Qin et al., 2023), and unsupervised methods have been applied directly to patient data, for example K-means clustering of exacerbation symptoms (Gill et al., 2024). This dissertation differs in its target. Instead of predicting a categorical pathogenicity call or an abstract deleteriousness score, it regresses real, continuously measured phenotypes, and it models the diploid genotype explicitly as a combination of two alleles.

V. Data and Methods

A. Dataset and pre-processing

Every analysis uses the phenotype summary table of Sosnay et al. (2013), which characterises 159 CFTR variants across roughly 39,696 patients and covers the majority of CF alleles by population frequency. The source table uses a merged two-row header, records start on the third row, and several trailing columns are empty; parsing left thirty-one columns, most of them metadata. The clinical outcomes are mean sweat chloride (mmol/L), pancreatic insufficiency (%), mean FEV1 (% predicted) and *Pseudomonas aeruginosa* infection (%). The functional features are variant type, chloride conductance (% of wild type), C/(B+C) protein processing in HeLa and FRT cells, and the proportion of correctly spliced RNA. The computational features are the POSE score, a CFTR-specific estimate supplied with the source dataset (Sosnay et al., 2013), together with PolyPhen-2, SIFT and PANTHER (Adzhubei et al., 2010; Kumar et al., 2009; Mi et al., 2013). Of the 159 variants, 127 were called CF-causing, 20 Indeterminate and 12 Not CF-causing. Three variants lack at least one clinical outcome, so every model and every clustering result below rests on the 156 variants with complete targets.

Variant names were cleaned of whitespace and non-breaking-space characters, the legacy and HGVS naming schemes were reconciled so that records matched consistently, and numeric columns were coerced to floating-point values. Missing data were handled deliberately. The functional features are absent for about 60% of variants, but this is expected, and not a fault in the data. Assays of channel conductance and protein processing are only meaningful for the 65 missense variants. For frameshift, nonsense and splice variants, function is abolished by mechanisms these assays do not measure. Coverage within the missense set is not perfectly uniform either, each assay having been quantified for a slightly different subset of 62 to 67 records. Missing values in the continuous functional features were filled with the feature median. Spliced RNA was a special case because it is left-censored, 80 values being recorded only as "<10" and imputed at a midpoint of 5; it was nonetheless left out of the feature set, because once filled in it is close to constant and it is never measured on the same variants as the missense-specific assays. Variant type was one-hot encoded across its six categories, the clinical call was encoded ordinally, and the ordinal PolyPhen-2 prediction was mapped to integers rising with predicted severity. The single-allele feature set therefore comprises thirteen features: the six variant-type indicators, chloride conductance, the two protein-processing measures and the four in-silico scores.

B. Models

The modelling proceeded in three stages. In the unsupervised stage, features were standardised, then reduced in

dimension by principal component analysis (PCA) and, separately, by uniform manifold approximation and projection (UMAP) (McInnes et al., 2018). Two feature sets were used in parallel: all 156 variants on the four clinical outcomes, the only features available for every variant, and the 65 missense variants on the full set of clinical, functional and in-silico features. Cluster structure was examined with K-means for two to eight clusters and checked against Ward hierarchical agglomerative clustering (Ward, 1963).

For the supervised single-allele stage, a multi-output Random Forest regressor (Breiman, 2001; Pedregosa et al., 2011) was trained to predict mean sweat chloride, pancreatic insufficiency and mean FEV1 together from the functional and in-silico features and the encoded variant type. *Pseudomonas* infection was retained for the exploratory and clustering analyses but excluded as a supervised target, being the noisiest of the four clinical variables and the most weakly associated with the molecular features. Random Forest was pre-specified as the primary model before any comparison was run: it handles mixed categorical and continuous inputs on very different scales, captures the non-linear interactions expected between variant type and functional impact, and stays stable at this sample size without the tuning that so few examples could not justify. It used the default scikit-learn configuration of one hundred trees grown until every leaf was pure, under a fixed random seed, and every family below was likewise run at its package defaults, so no model received more tuning effort than another.

The genotype-pair stage needed a way to represent two alleles at once without regard to their order. For each pair of variants, and for every feature, the minimum and the maximum of the two allelic values were computed, and the encoding was checked to confirm that exchanging the two alleles leaves it unchanged. Pair-level targets followed a minimum-function rule: pair sweat chloride and pancreatic insufficiency were taken as the minimum of the two allelic values and pair FEV1 as the maximum, reflecting the recessive expectation that the less severely affected allele governs residual function. A Random Forest was then trained on this pair representation. Finally, to establish that the findings are properties of the data and not artefacts of one algorithm's inductive bias, both stages were refitted with CatBoost, a gradient-boosted decision-tree method (Prokhorenkova et al., 2018), and with TabPFN, a transformer pre-trained on synthetic tabular data (Hollmann et al., 2022). Regression is supported only from version 2 of TabPFN onward, so the version 2 weights were used throughout (Hollmann et al., 2025). The three families carry very different inductive biases, so agreement between them is good evidence that a finding reflects the data.

C. Evaluation and leakage control

The single-allele model was evaluated by leave-one-out cross-validation (LOOCV) over the 156 variants with complete clinical targets: LOOCV maximises the data available for fitting on a small sample while still giving a fully out-of-sample estimate for every variant.

The genotype-pair model required more care. Because each variant appears in many different pairs, a plain random split would place pairs sharing a common allele in both the training and the test sets; the model could then memorise that allele's values and appear to predict well when in fact information had leaked across the split. This is a standard instance of leakage, in which information about the test set reaches the model through a route the analyst did not intend (Kaufman et al., 2012), and the accepted remedy for grouped observations is to split on the grouping variable rather than on the individual observation (Roberts et al., 2017). Pairs were therefore divided using variant-grouped 5-fold cross-validation (GroupKFold), which keys the split on the constituent variants so that every pair containing a given allele is confined to a single fold. Because no allele is ever split across training and test, the model cannot memorise an allele's clinical values and recognise them at test time, and is forced to generalise to allele combinations it has genuinely never seen. The 156 variants generate 12,246 unique unordered pairs. In addition, three clinically prominent variants (F508del, G551D and R117H) were withheld from training entirely, removing 465 pairs and leaving 11,781 for fitting, which provided a genuine out-of-sample test against their measured values. Each family was evaluated under this same protocol, with two exceptions: TabPFN's single-allele results used 5-fold cross-validation rather than LOOCV, and, because TabPFN accepts only a limited number of training rows, at the pair level it was fitted on a capped subsample while still being tested on held-out variants under the grouped scheme.

Performance was measured by the coefficient of determination (R^2), the root-mean-square error (RMSE) and the mean absolute error (MAE). Feature contributions were assessed by permutation importance, which records how far R^2 falls when a feature's values are shuffled at random and so stays robust to feature scale and to correlations between predictors. Cluster quality was summarised by the silhouette score (Rousseeuw, 1987), with the correspondence between discovered clusters and the known classification examined by cross-tabulation and by profiling mean clinical values within each cluster.

VI. Results

Results are presented with the supervised findings first, since they carry the headline claims; the unsupervised analysis that preceded them methodologically is reported in Section VI-D as a structural check. Before any modelling, the variables were characterised by classification. The four clinical outcomes separated clearly along the clinical call (Fig. 1): CF-causing variants carried the highest sweat chloride and pancreatic insufficiency and the lowest FEV1, while *Pseudomonas* infection was the noisiest variable, which is why it was not carried forward as a supervised target. Among the missense variants, chloride conductance and protein processing separated the classes as well.

A. Single-allele prediction

Evaluated by LOOCV, the single-allele Random Forest predicted pancreatic insufficiency and mean sweat chloride with moderate accuracy, while mean FEV1 remained effectively

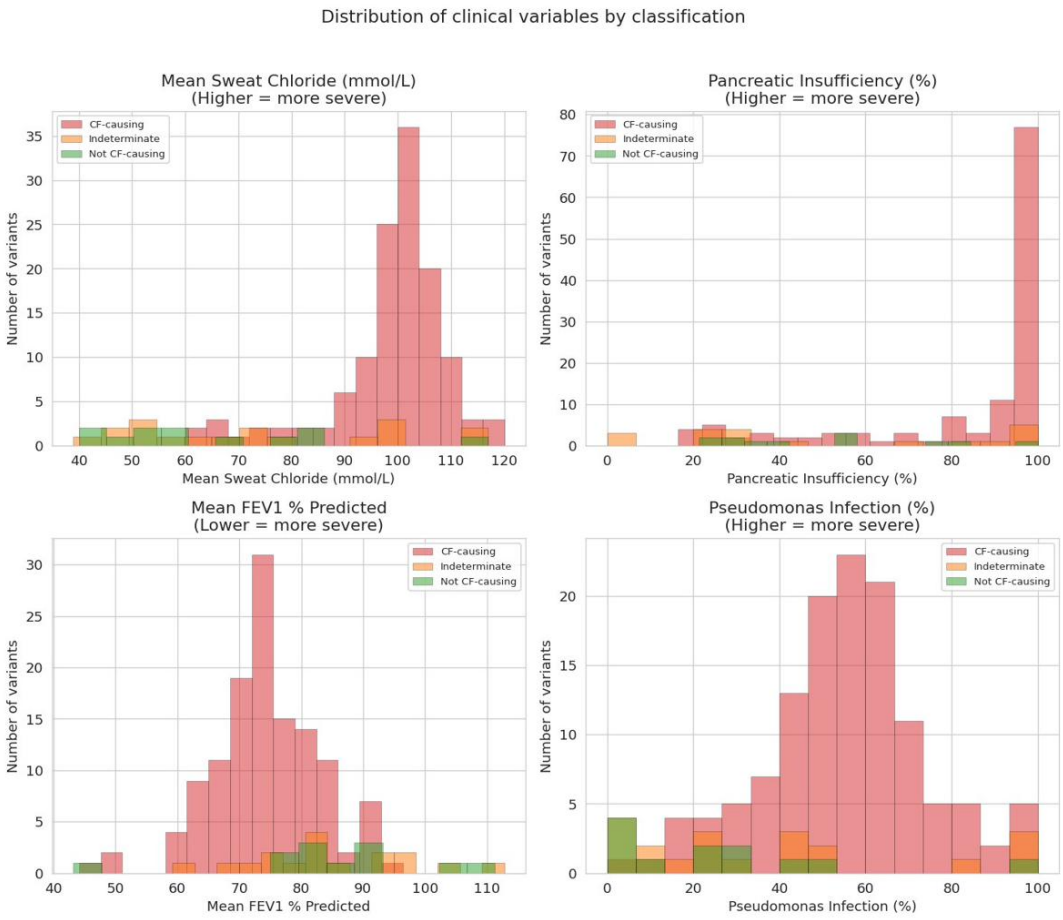


Fig. 1. Distribution of the four clinical variables by classification. Sweat chloride and pancreatic insufficiency separate the classes most cleanly, FEV1 shows moderate overlap, and *Pseudomonas* infection is the noisiest variable.

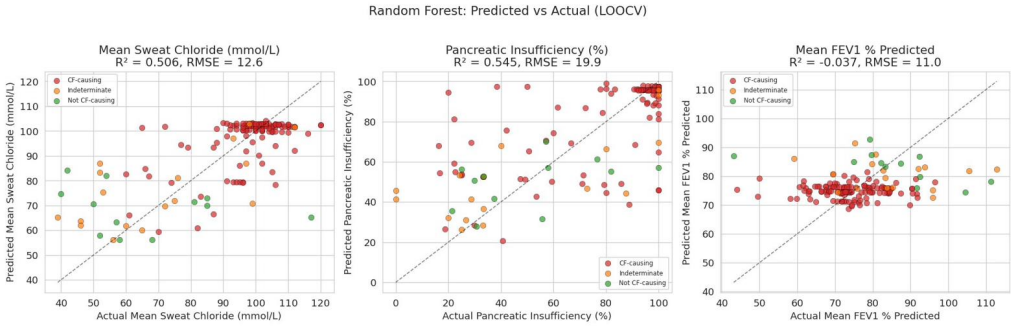


Fig. 2. Single-allele Random Forest predictions versus measured clinical values under leave-one-out cross-validation, coloured by clinical classification. Pancreatic insufficiency and sweat chloride are predicted with moderate accuracy; FEV1 is not.

unpredictable (Fig. 2). Pancreatic insufficiency was the best-predicted outcome ($R^2 = 0.545$), with sweat chloride close behind ($R^2 = 0.506$). For FEV1 the model could not even beat the mean, returning a slightly negative R^2 of -0.037. Residuals showed no pronounced systematic bias across the predicted range for any target, although they spread more widely in the sparsely populated, milder region of the sweat-chloride scale.

B. Genotype pairs, leakage and held-out validation

The genotype-pair model, evaluated under variant-grouped cross-validation, came in uniformly below the single-allele model on all three targets (Table I). That fall is the central quantitative result, and it needs explaining, because at first it looked very different. There is a structural reason to expect no improvement: the pair-level targets are deterministic functions of the single-allele values, obtained through the minimum and maximum rule, so pairing introduces no biological information

TABLE I
PREDICTIVE PERFORMANCE OF THE THREE MODEL FAMILIES ON THE SINGLE-ALLELE AND GENOTYPE-PAIR TASKS. RMSE AND MAE ARE IN THE UNITS OF THE TARGET (MMOL/L FOR SWEAT CHLORIDE, PERCENTAGE POINTS OTHERWISE). SINGLE-ALLELE RESULTS USE LEAVE-ONE-OUT CROSS-VALIDATION, EXCEPT TABPFN, WHICH USED 5-FOLD; GENOTYPE-PAIR RESULTS USE VARIANT-GROUPED (GROUPKFOLD) CROSS-VALIDATION.

		Single-allele			Genotype-pair		
Target	Model	R^2	RMSE	MAE	R^2	RMSE	MAE
Sweat chloride (mmol/L)	Random Forest	0.506	12.6	8.7	0.328	16.0	11.8
	CatBoost	0.512	12.5	8.7	0.491	14.0	10.5
	TabPFN	0.546	12.1	8.4	0.382	15.4	11.4
Pancreatic insufficiency (%)	Random Forest	0.545	19.9	13.1	0.271	27.2	19.6
	CatBoost	0.573	19.3	13.1	0.442	23.8	17.6
	TabPFN	0.551	19.8	13.6	0.465	23.3	16.6
FEV1 (% predicted)	Random Forest	−0.037	11.0	8.1	−0.495	12.0	9.0
	CatBoost	−0.028	11.0	8.2	−0.092	10.3	7.9
	TabPFN	−0.004	10.8	7.8	−0.210	10.8	8.1

TABLE II
HELD-OUT VALIDATION OF THE GENOTYPE-PAIR MODEL. F508DEL, G551D AND R117H WERE WITHHELD FROM TRAINING ENTIRELY, REMOVING 465 PAIRS AND LEAVING 11,781 FOR FITTING. THE TARGET IS THE PAIR-LEVEL VALUE IMPLIED BY THE MINIMUM-FUNCTION RULE APPLIED TO THE MEASURED ALLELIC SWEAT CHLORIDE IN THE SOURCE TABLE (SOSNAY ET AL., 2013), NAMELY 102 MMOL/L FOR F508DEL, 104 FOR G551D AND 60 FOR R117H. ALL THREE FAMILIES RECOVER THE CORRECT SEVERITY ORDERING. ALL THREE ALSO UNDER-PREDICT THE SEVERE GENOTYPES BY ROUGHLY 5 MMOL/L AND OVER-PREDICT THE R117H GENOTYPES BY ROUGHLY 7 MMOL/L, A COMPRESSION TOWARD THE CENTRE OF THE TRAINING DISTRIBUTION.

	Predicted (mmol/L)			Target	Expected
Genotype	RF	CatBoost	TabPFN	(mmol/L)	severity
F508del/F508del	97.1	95.7	94.5	102	Severe
F508del/G551D	96.7	98.5	94.8	102	Severe
G551D/G551D	98.7	101.0	95.4	104	Severe
F508del/R117H	67.5	70.6	65.0	60	Mild/mod.
R117H/R117H	67.4	64.5	62.4	60	Mild

the single-allele model did not already have. Pairing is useful for turning the model into a genotype-level lookup, not for extracting extra signal. Earlier versions of the pair model, built before variant-grouped cross-validation was in place, reported much higher numbers, including a sweat-chloride R^2 of about 0.85 against the 0.328 obtained once leakage was controlled. That performance was an illusion: because individual alleles recur across many pairs, the model had been recovering memorised allele values instead of generalising. The leakage-controlled figures in Table I are therefore the ones reported and defended here.

The held-out validation confirmed that the model still ranks genotype severity in the correct order even where it cannot predict precisely (Table II). Predicted sweat chloride for F508del/F508del (97.1 mmol/L), F508del/G551D (96.7 mmol/L) and G551D/G551D (98.7 mmol/L) fell about 5 mmol/L below their targets of 102, 102 and 104 mmol/L, while the two R117H-containing genotypes were predicted at 67.5 and 67.4 mmol/L against a target of 60, an over-prediction of roughly 7.5 mmol/L. The errors run in opposite directions at the two ends of the scale and both point inward, so the model pulls severe genotypes down and mild genotypes up. Every one of them is well inside the pair-level MAE of 11.8 mmol/L, which means the held-out genotypes are predicted no worse than the grouped cross-validation would lead one to expect; the compression toward the centre is taken up in Section VII. Fig. 3 collects predictions across every pair of the twenty most frequent variants, the clinician-facing lookup offered as the translational output of the work: symmetric by construction, it recovers the expected severity gradient and shows R117H-containing genotypes as a distinct low-sweat-chloride band. Given a patient's two variants, it returns an expected sweat-chloride level at a glance.

C. Feature impact

The correlation structure of the features (Fig. 4) agrees with known biology. Sweat chloride and pancreatic insufficiency were strongly and positively correlated ($r = 0.78$), chloride conductance ran inversely to sweat chloride ($r = -0.48$), and the HeLa and FRT protein-processing measures were almost perfectly collinear ($r = 0.97$). The blank cells linking spliced RNA to the missense-specific features are a property of the data and not an error. No variant belongs to both the splice-quantified and the missense-only subsets, so those correlations are undefined.

Permutation importance was computed separately for each target (Fig. 5). Chloride conductance emerged as the dominant predictor for all three outcomes, and by a wide margin for sweat chloride. This is biologically reassuring, since conductance is the most direct in-vitro readout of CFTR channel function and sweat chloride its closest clinical counterpart, so the model's reliance on it acts as a form of internal validation. It does need reading alongside the measurement structure: conductance is observed only for the missense variants and imputed at the median for the rest, so its leading importance is really driven by the missense subset. For non-missense variants the feature is constant and tells the model nothing, and there the variant-type indicators carry the discriminative weight instead. The two protein-processing measures need reading jointly rather than separately, since at $r = 0.97$ permutation importance divides the credit between them and understates each. Beyond conductance, secondary contributions came from the splice-other indicator and from PANTHER, POSE and

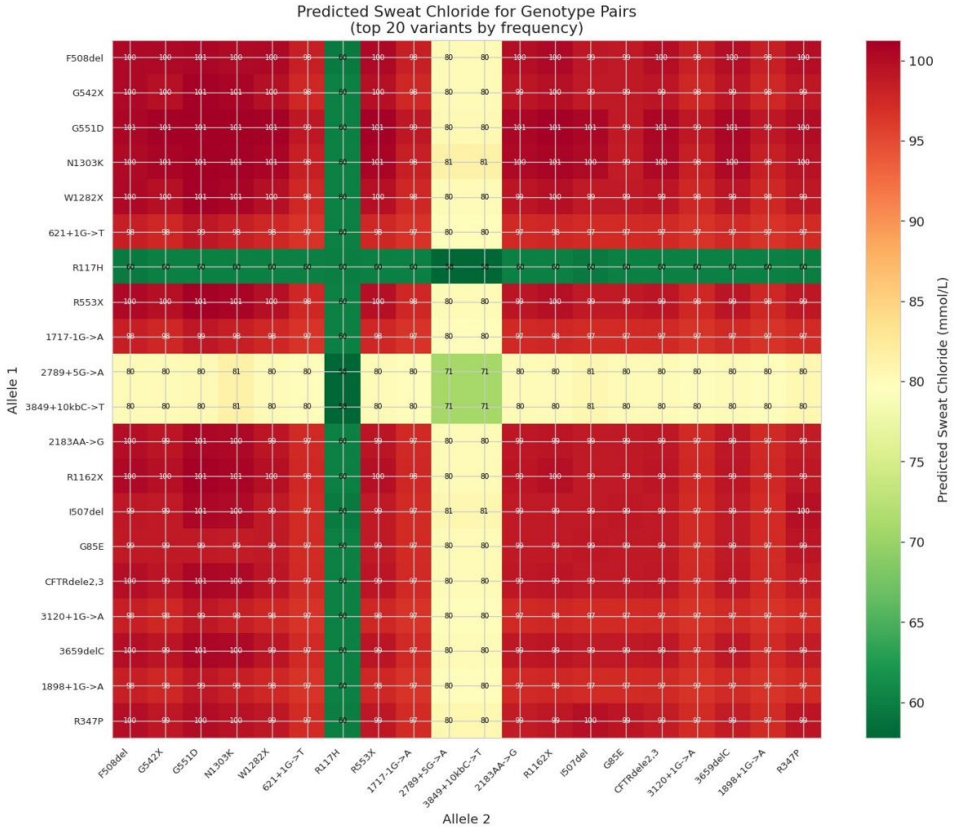


Fig. 3. Predicted mean sweat chloride across genotype pairs of the twenty most frequent CFTR variants. The matrix is symmetric by construction; R117H-containing genotypes form the characteristic low-sweat-chloride band. Values come from the full-data model and illustrate the severity gradient; the leakage-controlled estimates in Table I and the held-out validation in Table II are the quantitative results.

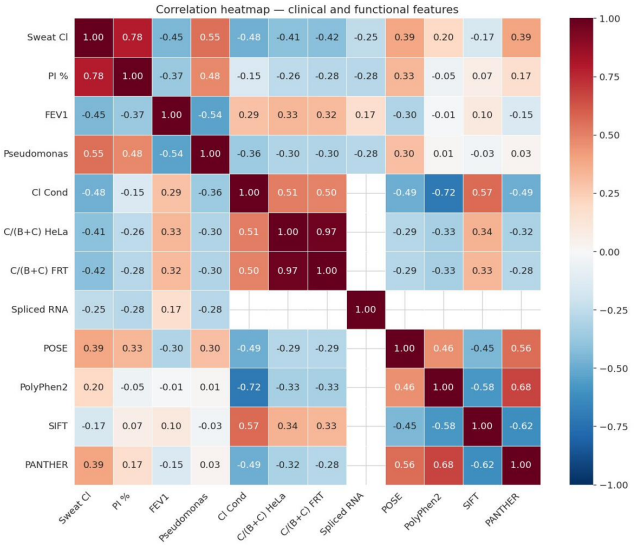


Fig. 4. Correlation structure of the clinical and functional features. Blank cells arise where two features are never measured on the same variant.

PolyPhen-2, and the ordering held up consistently across the three targets.

D. *Latent structure*

The unsupervised analysis confirmed that the data contain recoverable structure. On the clinical features across all variants, the first principal component accounted for 65.3% of the variance and the second for 18.6%, so two components together captured 83.9% (Fig. 6). This should not be over-interpreted: much of that concentration reflects the use of only four clinical inputs that are themselves mutually correlated, so the low apparent dimensionality is not necessarily a deep property of the phenotype. The leading component was driven by sweat chloride, pancreatic insufficiency and *Pseudomonas* infection acting against FEV1. On the missense subset the first two components capture less, 45.9% and 18.2% (Fig. 6, right). UMAP gave the same broad separation and is not shown.

K-means scored its highest silhouette at two clusters (0.507); at three the silhouette dropped to 0.273, yet that partition mapped far more interpretably onto the clinical classification. The $k = 2$ solution only separates the dense severe group from everything else, whereas $k = 3$ recovers the intermediate group needed to line up with the tripartite clinical call, so $k = 3$ was adopted for interpretation with the silhouette-optimal $k = 2$ reported for completeness. Cross-tabulation confirmed that the discovered clusters partly recover the known categories (Table III): clusters 0 and 2 are almost purely CF-causing, at 39 of 43 and 74 of 78 variants, whereas cluster 1 is mixed, holding 14 CF-causing variants against 12

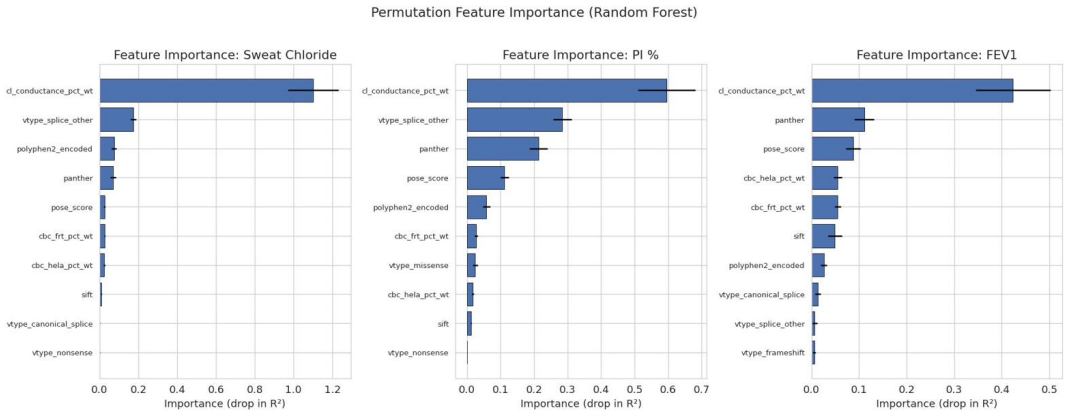


Fig. 5. Permutation feature importance (drop in R^2) for each clinical target. Chloride conductance dominates every target, most strongly for sweat chloride.

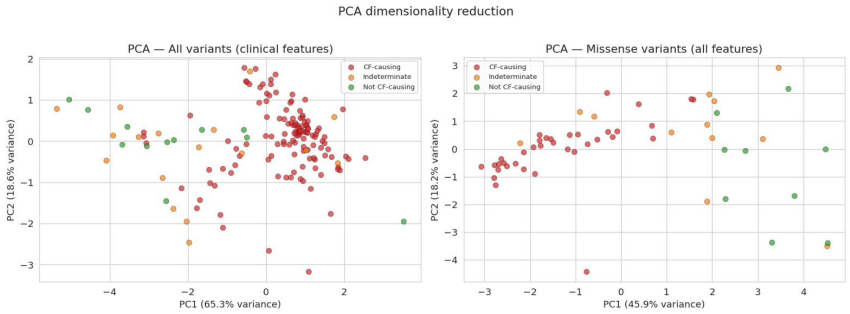


Fig. 6. PCA projection of the variants. Left: all 156 variants on clinical features, the first two components capturing 83.9% of variance. Right: missense variants on the full feature set. Points are coloured by clinical classification.

TABLE III

PROFILE OF THE THREE K-MEANS CLUSTERS ($k = 3$) OVER THE 156 VARIANTS WITH COMPLETE CLINICAL TARGETS: CLUSTER SIZE, COMPOSITION BY CLINICAL CALL, AND MEAN MEASURED AND PREDICTED VALUE OF EACH OUTCOME. CLUSTERS 0 AND 2 ARE SEVERE, CLUSTER 1 MILD. PREDICTIONS ARE COMPRESSED TOWARD THE CENTRE, MOST VISIBLY FOR THE MILD CLUSTER, AS EXPECTED WHEN ROUGHLY FOUR FIFTHS OF VARIANTS ARE CF-CAUSING.

		Clinical call			Sweat chloride (mmol/L)		PI (%)		FEV1 (% pred.)	
Cluster	n	CF-causing	Indet.	Not CF	Measured	Predicted	Measured	Predicted	Measured	Predicted
0 (severe)	43	39	3	1	101.4	96.1	90.3	82.3	66.7	75.3
1 (mild)	35	14	12	9	64.2	75.9	31.2	49.7	85.5	79.7
2 (severe)	78	74	2	2	100.8	99.0	91.1	88.9	77.1	75.0

Indeterminate and 9 Not-CF-causing. Recovery is therefore strong at the severe end and weak at the mild end, which is what the class imbalance would predict. Ward hierarchical clustering produced a consistent picture, its primary split echoing the K-means division.

Profiling the single-allele model's predictions by cluster membership (Table III, right-hand columns) shows the mild cluster carrying the lowest predicted sweat chloride and pancreatic insufficiency and the highest predicted FEV1, with the severe clusters showing the reverse, so the ordering of the predictions reproduces the ordering of the measurements. The agreement is one of rank, though, and not of magnitude. The mild cluster is measured at 64.2 mmol/L but predicted at 75.9, and at 31.2% pancreatic insufficiency but predicted at 49.7%. Roughly four fifths of the variants are CF-causing, so the leaf averages of the Random Forest are dominated by severe values and the model is reluctant to predict far into the mild range. The clustering, which uses no labels and no model, is under no such pressure and separates the mild group cleanly. The two analyses therefore agree on which variants are severe while disagreeing on how mild the mild ones are, and it is the supervised model that is at fault on the second point. This is the same compression seen in the held-out genotypes of Table II, arrived at by an independent route.

E. Robustness across model families

Repeating both analyses with CatBoost and TabPFN does not change the picture (Table I). For the single-allele model, sweat chloride sits at an R^2 of about 0.51 for both Random Forest and CatBoost and 0.55 for TabPFN, pancreatic insufficiency ranges only from 0.55 to 0.57, and FEV1 stays just below zero for all three, at -0.037, -0.028 and -0.004. The

spread across families on any one target is about 0.04 in R^2, well within what a sample of 156 variants can resolve, so no model beats the others. Both headline findings survive the change of model.

The genotype-pair results tell much the same story with one wrinkle. CatBoost and TabPFN sit somewhat above Random Forest on sweat chloride and pancreatic insufficiency, CatBoost reaching $R^2 = 0.49$ for sweat chloride against the Random Forest's 0.33. This does not overturn the conclusion: even the strongest pair model does not exceed the single-allele figure of about 0.51, so pairing still buys no real gain, and FEV1 stays negative for every family at the pair level. The boosted and transformer-based models are simply a little more robust on the harder, leakage-controlled problem. The error metrics follow the same ordering, pancreatic insufficiency carrying the largest RMSE and MAE because its values run across the full range from 0 to 100.

VII. Discussion

A. The predictable and the unpredictable

The pattern of what the model can and cannot predict is itself the central finding, and it maps cleanly onto CF biology. Sweat chloride and pancreatic insufficiency sit most directly downstream of CFTR channel function: sweat chloride is essentially a direct readout of epithelial chloride transport, and exocrine pancreatic status is largely fixed early in life by the residual function of the two CFTR alleles. Because these phenotypes are close to the molecular defect, the functional assays and effect scores carry real information about them, and the model recovers that signal at moderate accuracy.

FEV1 behaves in the opposite way. Its near-zero, slightly negative R^2 is a biological result and not a modelling failure. Lung disease in CF is not fixed by the CFTR genotype; it accumulates over a lifetime under the influence of modifier genes at loci such as 11p13 and 22q12, chronic *Pseudomonas* infection, treatment adherence, nutrition and socioeconomic circumstance (Wright et al., 2011; Shanthikumar et al., 2019). None of these is encoded in a variant-level feature vector, so no model built on that vector, however flexible, can explain the between-patient variance in lung function; a model reporting a high R^2 for FEV1 from genotype alone would in fact be cause for suspicion. The near-zero result is the one consistent with decades of clinical observation, and it holds identically across all three families, which rules out an idiosyncrasy of Random Forest. The feature importances reinforce this reading of the model as biologically well-behaved, and the held-out genotypes are ranked correctly even though those variants were never seen in training.

B. Relationship to previous work

These results align with, and sharpen, a long line of CF research. The classical studies established that pancreatic status and sweat chloride separate severe from mild genotypes with reasonable consistency, while pulmonary outcomes resist genotype-based explanation (Kerem et al., 1990; Borgo et al., 1990; Wilschanski et al., 1995); this dissertation puts numbers on that ordering where the earlier work described it, attaching concrete predictive metrics to each phenotype. The comparison with recent machine-learning work is equally pointed: prognostic models for CF lose accuracy when transferred between registries (Qin et al., 2023), and unsupervised methods have recovered phenotypes directly from patient data (Gill et al., 2024). Both are consistent with what is found here, that the recoverable structure is real but modest and the honest ceiling lower than a naive analysis suggests. The distinctive contribution is methodological: pairing an order-invariant genotype encoding with strict variant-grouped validation shows precisely how allele memorisation manufactures inflated performance, and how much of it evaporates once the leakage is closed.

C. Limitations

The most fundamental limitation is the size and shape of the dataset. One hundred and fifty-six variants is a small sample for supervised learning: it limits model complexity, widens the uncertainty on every reported metric, and leaves the mild end of the severity spectrum thinly populated. That sparsity has a visible consequence, since predictions are compressed toward the centre of the training distribution at both ends of the scale: R117H/R117H is placed at 67.4 mmol/L against a target of 60, while F508del/F508del is placed at 97.1 against a target of 102. With few mild examples to learn from, and with Random Forest predictions bounded by the leaf averages of a severe-dominated training set, some regression toward the majority is unavoidable, and it is the same compression that appears in the cluster profiles of Table III.

Missing data impose a second constraint. The functional assays are recorded only for the missense subset, so for the roughly 60% of non-missense variants they were imputed at the median. This is defensible, because function is abolished by mechanisms these assays do not measure, but it does mean that for most variants the functional columns carry no discriminating information and the model leans on variant type instead. Third, the genotype-pair analysis rests on a modelling assumption: the minimum-function rule encodes the recessive expectation that the milder allele governs residual function, a reasonable first approximation that cannot represent dominant-negative behaviour or more complex allele interactions. Table II shows the cost directly. The rule assigns F508del/R117H and R117H/R117H the same target of 60 mmol/L, because both genotypes inherit R117H's allelic value, and the model duly returns 67.5 and 67.4 mmol/L for them. Clinically the two are not equivalent, and the severity of R117H itself depends on the poly-thymidine tract carried in cis with it (Kiesewetter et al., 1993), so no rule keyed on the milder allele alone can separate them. Because the pair targets are deterministic functions of the single-allele values, the pair model also inherits the single-allele error instead of reducing it. Fourth, a curated disease-variant database is enriched for CF-causing alleles and for variants common enough in largely European-ancestry cohorts to have been characterised, so both the class balance and the allele spectrum are skewed relative to the global patient population. A related caveat applies to the unsupervised validation, since the clinical variables used

for clustering also informed the original Sosnay classification, so the partial recovery of that classification is to some degree built in and is best read as an internal consistency check.

These constraints converge on a single caution about generalisation. All results derive from one dataset, external validity has not been tested, and CF prognostic models are known to degrade under exactly that kind of transfer (Qin et al., 2023). The pairwise lookup in Fig. 3 should therefore be treated as an illustrative severity gradient from the full-data model, not a clinically validated predictor; the leakage-controlled metrics in Table I and the held-out validation in Table II remain the defensible quantitative claims.

D. Future directions

The clearest extension is to move from cross-sectional to longitudinal modelling. FEV1 changes over a lifetime, and its rate of decline is what matters clinically; a model trained on repeated registry measurements could learn the shape of that decline and might recover signal invisible in a static snapshot, while also enlarging the sample well beyond a single cross-sectional table. A complementary direction is to supply the information the model currently lacks: genotypes at established modifier regions such as 11p13 and 22q12 alongside markers of infection status and treatment (Wright et al., 2011; Shanthikumar et al., 2019), and richer molecular descriptors such as structural or evolutionary embeddings of CFTR, so that non-missense variants are no longer represented almost entirely by their variant-type indicator. A cheaper immediate test is a composition baseline: predict each allele with the single-allele model and apply the same minimum rule to the two predictions, which would establish whether a dedicated pair model earns its place at all. Methodologically, the framework transfers directly, since the same order-invariant encoding and variant-grouped validation apply to other recessive disorders, and the family comparison extends to larger tabular foundation models as they mature.

VIII. Conclusion

This dissertation set out to determine which cystic fibrosis phenotypes can be predicted from CFTR variant features, under an evaluation strict enough to be believed. Sweat chloride and pancreatic insufficiency are moderately predictable ($R^2 = 0.506$ and 0.545) because they sit close to CFTR channel function, whereas FEV1 is not predictable from genotype alone ($R^2 \approx -0.04$) because lung disease is written mostly by modifiers, infection and environment. The genotype-pair extension delivered the sharpest methodological lesson: naive pairing appeared to reach a sweat-chloride R^2 near 0.85, but leakage-controlled, variant-grouped validation reduced this to an honest 0.328, exposing allele memorisation instead of genuine skill. That every result held across Random Forest, CatBoost and TabPFN confirms they are properties of the data, not of a single algorithm. The work leaves two contributions: a clinician-facing lookup of expected sweat chloride for any pair of the most common variants, offered as an illustrative severity gradient rather than a validated predictor, and a reusable, leakage-aware template for genotype-phenotype modelling that applies directly to other monogenic diseases.

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MSc Project - Reflective Essay

Project Title:	Genotype-Phenotype Mapping in Cystic Fibrosis: A Predictive Model of Disease Severity Based on Allelic Combinations
Student Name:	Elidian Terziu
Student Number:	250714281
Supervisor Name:	Zunaira Nadeem
Programme of Study:	Advanced Computer Science

What I did

I used one dataset, the supplementary table from Sosnay et al. (2013), which pulls together clinical and laboratory measurements for a set of CFTR variants.The work ran in two phases. First I looked at the data without any labels, using PCA, UMAP and k-means, to see whether the variants formed natural severity groups on their own. Then I trained a Random Forest to predict three clinical outcomes: sweat chloride, pancreatic insufficiency and lung function. Since real patients carry two alleles rather than one, I extended the model from single variants to pairs, and ran the same task through CatBoost and TabPFN to check the findings were not a quirk of one model.

Contribution and further work

The main contribution is about method rather than biology. When I first trained the pair model with an ordinary random split, the sweat chloride result looked very good. It was too good. Each allele appears in many different pairs, so a random split puts the same allele in training and in testing, and the model simply memorises it. Once I grouped the cross-validation by allele, so both alleles of a test pair were genuinely unseen, the score fell a long way. The lower figure is the honest one, and I report it as a conservative floor. Anyone building this kind of combination model from a short list of variants will hit the same trap.The second point is smaller but practical. My first pair encoding gave a different prediction depending on which allele was listed first. A genotype has no order, so I replaced those features with minimum and maximum values, which read the same either way round.The third is a negative result worth reporting. Lung function could not be predicted from variant properties at all. That fits what is already known, because lung function depends on modifier genes, infection history and treatment rather than on the variant itself.The obvious next step is patient-level data instead of variant averages, so individual variation can be modelled rather than averaged away. Adding known modifier genes as features would be a natural way to test why lung function fails, and the pair predictions would need checking against real genotype records before anyone treated them as more than a hypothesis.

Limitations and challenges

The dataset is small, and the laboratory features only exist for missense variants, so a lot of those columns are empty. That is not a data quality problem. A nonsense or frameshift variant makes no protein, so there is nothing to measure. It still limits what the model can learn.Every value is an average for a variant rather than a reading from one patient, so the model describes groups and not individuals. The data is also unbalanced, with most variants being severe, so predictions get pulled towards the severe end and the model is weakest on mild variants. That is exactly where a wrong answer would matter most.The unsupervised part is partly circular, since the same clinical variables were used both to build the clusters and to describe them. I say so in the dissertation rather than presenting the clusters as independent confirmation.Practically, cleaning the data took far longer than I planned. The source table has merged headers, hidden spaces inside variant names and censored values stored as text. I was also on the free tier of Google Colab, so long cross-validation runs kept losing their session and I had to restructure the notebook to save its progress and resume.

Personal development

The most useful thing I learned is to be suspicious of my own good results. I accepted that first high score for longer than I should have, mainly because it was the answer I wanted. Now I set the validation up properly before I look at any score.I also learned to report a result I did not like. Keeping the failed lung function model in, and explaining why that failure is the right outcome, was harder than deleting it would have been. The same goes for keeping Random Forest as the main model when two others scored higher on the pair task. It was the model I set out to use, and swapping it afterwards would have been selective reporting.Finally, I am much more comfortable with the biology than I was. I had no background in genetics before this, and being able to explain why sweat chloride is predictable and lung function is not is the part I am most pleased with.

ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: Elidian Terziu
Project Title: Genotype-Phenotype Mapping in Cystic Fibrosis: A Predictive Model of Disease Severity Based on Allelic Combinations

Supervisor Name: Zunaira Nadeem

1. Have you used Generative AI tools for your project?

YES

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
e.g. ChatGPT	5.6	OpenAI	https://chat.openai.com/

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Example Table – please add your own

Project Section	Description	Justification
e.g. Literature Review	Rephrased a sentence on ethical AI frameworks	I used GenAI to improve clarity but verified content against journal articles
Research	It gave me papers/journals to read	It gave me several papers that would help me to work on my thesis to read.
coding	Helped me interpret error messages	Helped me interpret error messages debug scripts and suggestions of library functions

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed:  Date: 14/08/2026

Guidance notes for students

- Students are strongly encouraged to review the following resources
    - Student Guide to Gen AI https://www.qmul.ac.uk/library/academic-skills/student-guide-to-generative-ai/
    - Academic Integrity and Misconduct Policy https://www.qmul.ac.uk/governance-and-legal-services/media/dgls-media/policy/current-policies/Academic-Integrity-&-Misconduct-Policy.pdf

- The use of essay mills, homework help sites, plagiarism, collusion, falsification, impersonation, or actions that may give an unfair advantage is strictly prohibited in your final year project. Please refer to the QMUL Academic Integrity and Misconduct policy above.
- It is never acceptable to use GenAI tools to write your entire literature review and submit it as your own work; doing so is considered a form of plagiarism since GenAI outputs are derived from ideas of human authors without proper referencing.
- You may use GenAI tools for checking grammar and spelling, provided the tool is not rewriting a large portion of your work for your final year project.
- Using GenAI for idea generation or initial planning is generally appropriate but always consider the context and assessment requirements for your project. If in doubt, please talk to your supervisor.
- If you are ever unsure whether GenAI is permitted for any part of your project always ask your project supervisor for explicit clarification.
- Unless stated otherwise, GenAI should only be used to assist your learning at the initial stages of your project and your final submission of your project, and the code must still be substantially your own original work.
- Inappropriate use of GenAI will be viewed as plagiarism under the QMUL academic integrity and misconduct policy this includes submitting work where GenAI's contribution exceeds what is reasonably allowed in the final year project.
- Attempting to pass off GenAI-generated content as your own can result in disciplinary investigation under academic misconduct regulations; you may be asked to attend a viva to confirm authorship.
- Before using GenAI, make sure:
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- Always consult and cite original, credible sources; GenAI should never be treated as an original source and should not be cited as an author or in the reference list.
- Critically assess all GenAI output for accuracy, double-check any facts, quotations, or citations it suggests especially if you use them in your literature review or other chapters.
- If GenAI has been used for drafting, planning, or structuring, you must clearly acknowledge and document its use in your submission. Therefore, you should keep drafts and records of your own process to show your independent scholarly work and reading notes
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    - Summarising information to prepare for more in-depth research.
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- Please note that output from GenAI is typically treated as having no author so you cannot cite Gen AI as an author in your report
- When in doubt, transparency and honesty about your methods and sources including GenAI are expected and protect your academic integrity